BROWN REPRESENTABILITY FOR COMPACTLY GENERATED TRIANGULATED CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The compact-generator cellular approximation lemma is included with its complete proof, not granted for free and not counted separately.

Editorial note. This is the compactly generated theorem. The broader brown-bis claim and its unresolved proof issue are not used. Original attribution to Neeman is retained.

Editorial scope. One original target with its local construction proved in full. Classical category primitives are explicitly allowed.

Allowed standard prerequisites (editorial).

- Yoneda lemma and cohomological functors. An element of H(X) gives a natural map Hom(-,X)→H; exact triangles yield long exact sequences.
- Compact objects and countable mapping telescopes. A compact C has Hom(C,hocolim X_n)=colim Hom(C,X_n). A compact generating family detects zero objects through all shifts.
- Localizing closure of the isomorphism locus. The locus where a natural transformation of cohomological functors is an isomorphism on every shift is triangulated and closed under summands; if both preserve coproducts as products it is closed under coproducts.

Source reference. Original lemma 001P: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 540–558. Standard Yoneda lemma. Stable Stacks tag displayed.

Source reference. Original lemma 094A: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 10405–10422. Explicit compactness/telescope identity permitted as standard. Stable Stacks tag displayed.

Source reference. Original lemma 05RD: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 1812–1825. Standard long-exact-sequence closure verification, not representability. Stable Stacks tag displayed.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category with direct sums which is compactly generated. Let $H : \mathcal{D} \to Ab$ be a contravariant cohomological functor which transforms direct sums into products. Then H is representable.*

Lemma 0.2. *Let $\mathcal{D}$ be a triangulated category with direct sums. Let E_i, $i \in I$ be a family of compact objects of $\mathcal{D}$ such that $\bigoplus E_i$ generates $\mathcal{D}$. Then every object X of $\mathcal{D}$ can be written as*

$$X = hocolim X_n$$

where X_1 is a direct sum of shifts of the E_i and each transition morphism fits into a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ where Y_n is a direct sum of shifts of the E_i.

Date: stacks-project:derived.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Set $X_1 = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathbf{Z}$ and $\varphi : E_i[m] \to X$. Then X_1 comes equipped with a canonical morphism $X_1 \to X$. Given $X_n \to X$ we set $Y_n = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathbf{Z}$, and $\varphi : E_i[m] \to X_n$ is a morphism such that $E_i[m] \to X_n \to X$ is zero. Choose a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ and let $X_{n+1} \to X$ be any morphism such that $X_n \to X_{n+1} \to X$ is the given one; such a morphism exists by our choice of Y_n. We obtain a morphism $\mathrm{hocolim} X_n \to X$ by the construction of our maps $X_n \to X$. Choose a distinguished triangle

$$C \to \mathrm{hocolim} X_n \to X \to C[1]$$

Let $E_i[m] \to C$ be a morphism. Since E_i is compact, Lemma 094A implies the composition $E_i[m] \to C \to \mathrm{hocolim} X_n$ factors through X_n for some n, say by $E_i[m] \to X_n$. Then the construction of Y_n shows that the composition $E_i[m] \to X_n \to X_{n+1}$ is zero. In other words, the composition $E_i[m] \to C \to \mathrm{hocolim} X_n$ is zero. This means that our morphism $E_i[m] \to C$ comes from a morphism $E_i[m] \to X[-1]$. The construction of X_1 then shows that such morphism lifts to $\mathrm{hocolim} X_n$ and we conclude that our morphism $E_i[m] \to C$ is zero. The assumption that $\bigoplus E_i$ generates $\mathcal{D}$ implies that C is zero and the proof is done. □

Proof. Let E_i, $i \in I$ be a set of compact objects such that $\bigoplus_{i \in I} E_i$ generates $\mathcal{D}$. We may and do assume that the set of objects $\{E_i\}$ is preserved under shifts. Consider pairs (i, a) where $i \in I$ and $a \in H(E_i)$ and set

$$X_1 = \bigoplus\nolimits_{(i,a)} E_i$$

Since $H(X_1) = \prod_{(i,a)} H(E_i)$ we see that $(a)_{(i,a)}$ defines an element $a_1 \in H(X_1)$. Set $H_1 = \mathrm{Hom}_{\mathcal{D}}(-, X_1)$. By Yoneda's lemma (Categories, Lemma 001P) the element a_1 defines a natural transformation $H_1 \to H$.

We are going to inductively construct X_n and transformations $a_n : H_n \to H$ where $H_n = \mathrm{Hom}_{\mathcal{D}}(-, X_n)$. Namely, we apply the procedure above to the functor $\mathrm{Ker}(H_n \to H)$ to get an object

$$K_{n+1} = \bigoplus\nolimits_{(i,k),\ k \in \mathrm{Ker}(H_n(E_i) \to H(E_i))} E_i$$

and a transformation $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to \mathrm{Ker}(H_n \to H)$. By Yoneda's lemma the composition $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to H_n$ gives a morphism $K_{n+1} \to X_n$. We choose a distinguished triangle

$$K_{n+1} \to X_n \to X_{n+1} \to K_{n+1}[1]$$

in $\mathcal{D}$. The element $a_n \in H(X_n)$ maps to zero in $H(K_{n+1})$ by construction. Since H is cohomological we can lift it to an element $a_{n+1} \in H(X_{n+1})$.

We claim that $X = \mathrm{hocolim} X_n$ represents H. Applying H to the defining distinguished triangle

$$\bigoplus X_n \to \bigoplus X_n \to X \to \bigoplus X_n[1]$$

we obtain an exact sequence

$$\prod H(X_n) \leftarrow \prod H(X_n) \leftarrow H(X)$$

Thus there exists an element $a \in H(X)$ mapping to (a_n) in $\prod H(X_n)$. Hence a natural transformation $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ such that

$$\mathrm{Hom}_{\mathcal{D}}(-, X_1) \to \mathrm{Hom}_{\mathcal{D}}(-, X_2) \to \mathrm{Hom}_{\mathcal{D}}(-, X_3) \to \ldots \to \mathrm{Hom}_{\mathcal{D}}(-, X) \to H$$

commutes. For each i the map $\mathrm{Hom}_{\mathcal{D}}(E_i, X) \to H(E_i)$ is surjective, by construction of X_1. On the other hand, by construction of $X_n \to X_{n+1}$ the kernel of

$\mathrm{Hom}_{\mathcal{D}}(E_i, X_n) \to H(E_i)$ is killed by the map $\mathrm{Hom}_{\mathcal{D}}(E_i, X_n) \to \mathrm{Hom}_{\mathcal{D}}(E_i, X_{n+1})$. Since

$$\mathrm{Hom}_{\mathcal{D}}(E_i, X) = \mathrm{colim}\, \mathrm{Hom}_{\mathcal{D}}(E_i, X_n)$$

by Lemma 094A we see that $\mathrm{Hom}_{\mathcal{D}}(E_i, X) \to H(E_i)$ is injective.

To finish the proof, consider the subcategory

$$\mathcal{D}' = \{Y \in \mathrm{Ob}(\mathcal{D}) \mid \mathrm{Hom}_{\mathcal{D}}(Y[n], X) \to H(Y[n]) \text{ is an isomorphism for all } n\}$$

As $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ is a transformation between cohomological functors, the subcategory $\mathcal{D}'$ is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ (details omitted; see proof of Lemma 05RD). Moreover, as both H and $\mathrm{Hom}_{\mathcal{D}}(-, X)$ transform direct sums into products, we see that direct sums of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Thus derived colimits of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Since $\{E_i\}$ is preserved under shifts, we see that E_i is an object of $\mathcal{D}'$ for all i. It follows from Lemma 0.2 that $\mathcal{D}' = \mathcal{D}$ and the proof is complete. □